

BGA School Building Project

Architectural Design Report

A G+2 Campus Plan for Integrated Learning



Team: Subhraneel Bag (24CE10111) · Shrehan Raj Singh (24CE10103) · Sk Toufik Haque (24CE10106) · Ayan Kumar (24CE10037) · Khade Aditya Santosh (24CE10070) · Ayush Nayal (24CE10040) · Rushil Ajaria (24CE10094) · Shourya Vinod Singh (24CE10002) · Hiya Agarwal (24CE10061) · Amish Khare (24CE10029) · Arsh M. Rashania (24CE10034) · Abhay (24CE10023)

1. INTRODUCTION

This report presents a comprehensive architectural analysis of the **BGA School Building Project** - a meticulously planned G+2 educational campus designed to support academic excellence, extracurricular development, and safe, efficient campus operations from Nursery through Class XII.

The project provides a structured view of the school building design, with each floor plan presented alongside detailed spatial descriptions, dimensional data, and 3D visualizations developed in Blender 4.3.2. The design integrates classrooms, administrative facilities, science laboratories, a library, sports infrastructure, and landscaped open spaces within a unified campus master plan.

1.1 Project Objectives

- Design a **G+2 school building** with doubly loaded corridors for efficient circulation
- Accommodate **Nursery to Class XII** with age-appropriate spatial zoning across three floors
- Integrate **sports facilities** - cricket ground, football ground, basketball court, badminton court, and children's play area
- Ensure **safe entry and exit** points with segregated parking for school buses, staff, and visitors
- Provide **accessible design** with ramps, elevators, and wide doorways

1.2 Design Philosophy

The campus is organized around a **central courtyard** that serves as the heart of the building - bringing natural light and ventilation into doubly loaded corridors while hosting a badminton court for recreational use. Academic zones are distributed by age group vertically: primary classes on the ground floor, secondary classes on the first floor, and specialized labs and creative spaces on the second floor.

Design Highlight: The doubly loaded corridor typology maximizes classroom frontage along circulation spines, while the central courtyard creates a visual and environmental focal point - a strategy seen in successful Indian campus designs such as Vidyagyan School, Sitapur, where academic blocks cluster around central courts.

1.3 Building Program Overview

Academic Zone	Admin & Support	Sports & Recreation
Nursery through Class XII classrooms (7 m × 7 m each), science labs, computer lab, library, music room, and seminar/conference facilities across three floors.	Principal's office, general office, parents' waiting area, sick room, store, sports kits room, counter, and teachers' lounge.	Cricket ground, football ground, basketball court, courtyard badminton court, children's play area, and landscaped gardens.

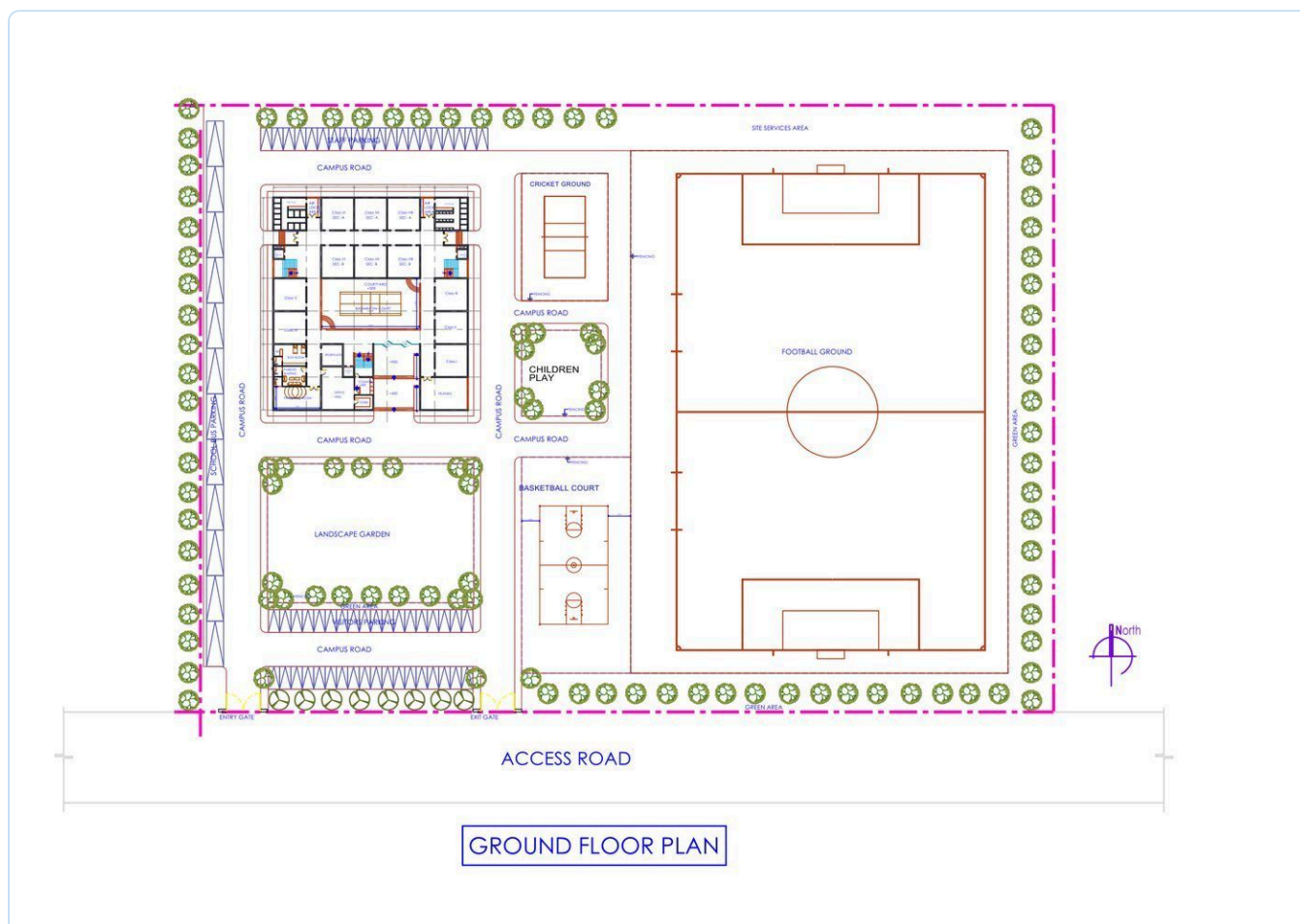
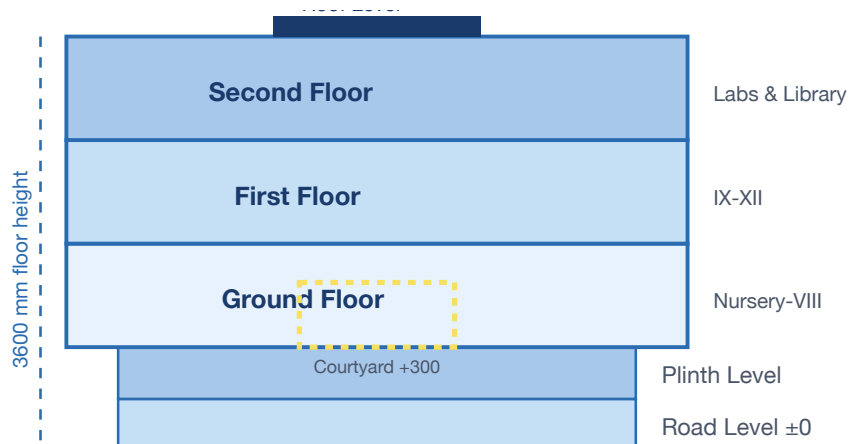


Fig 1: Campus bird's-eye overview showing the G+2 school building, sports fields, parking zones, and landscaped green areas.

1.4 Vertical Building Section



2. CAMPUS MASTER PLAN

The complete school layout ensures a well-structured learning environment with clear zoning between academic, administrative, sports, and service areas. The campus is accessed via a primary **entry gate** and **exit gate** along an access road, with internal campus roads (7250 mm width) connecting all zones.

2.1 Site Zoning

Sports Zone: Cricket ground (east), football ground (west), basketball court, and children's play area with fencing. The courtyard badminton court (21.5 m × 11 m) is integrated within the building footprint at +300 mm level.

Parking: Segregated zones for school bus parking, staff parking, and visitors parking - ensuring safe drop-off and pick-up operations.

Green Areas: Landscape gardens, green buffers along campus roads, and fenced green areas providing visual relief and environmental quality.

Services: Site services area, plumbing pipe routes, and electrical ducts planned along circulation spines for maintainability.

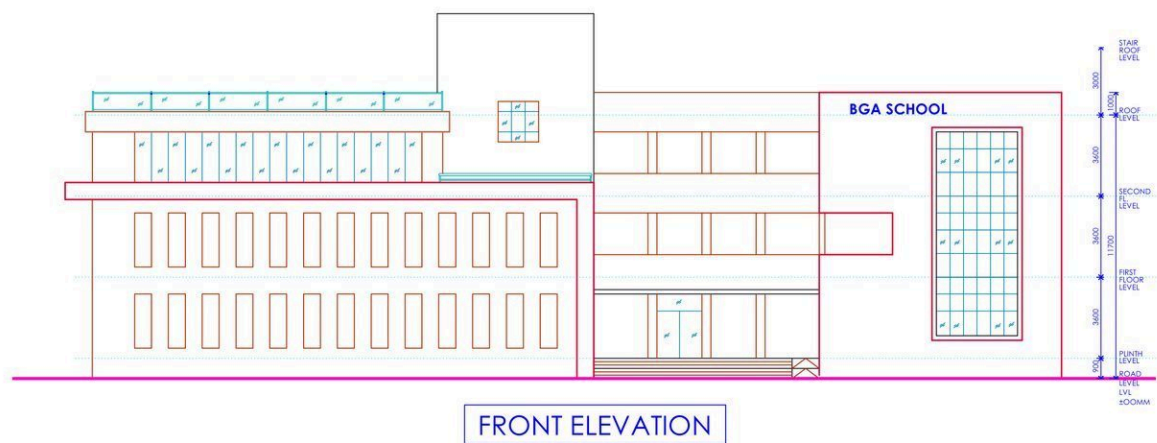


Fig 3: Annotated ground floor site plan showing classroom wings, courtyard, sports facilities, parking, and campus roads with dimensional annotations (mm).

2.2 Circulation Flow

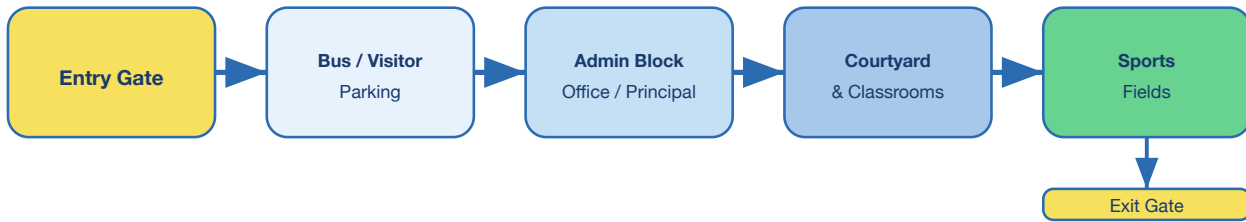


Fig 4: Campus circulation flow - entry gate → parking → administration → courtyard/classrooms → sports fields → exit gate.

2.3 Key Campus Dimensions

Element	Dimension	Notes
Campus roads	7250 mm width	Internal vehicular and pedestrian circulation
Courtyard / Badminton	21.5 m × 11 m	Central recreational court at +300 mm level
Children's play area	Adjacent to admin block	Fenced for safety
Centre table (play equipment)	1000 × 500 × 400 mm	Two units in play zones
Building footprint	G+2 structure	Doubly loaded corridors, 3600 mm floor heights



Fig 5: Campus plan - upper floors and site context.

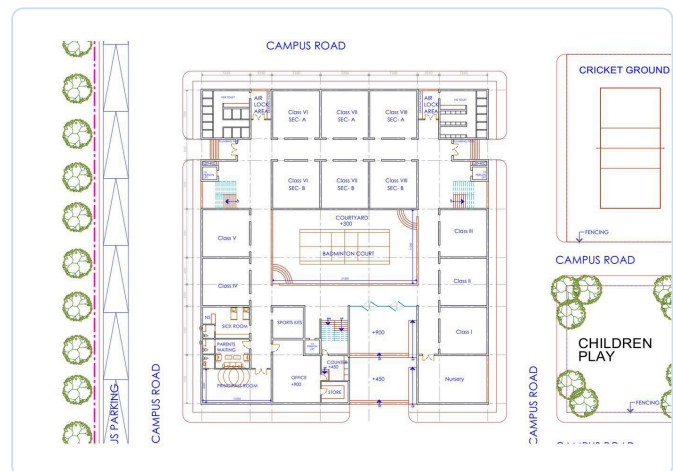
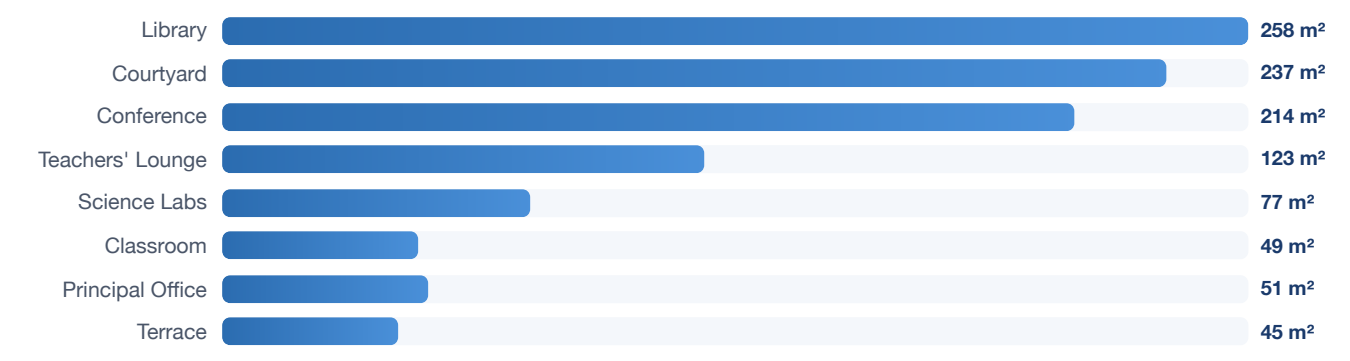


Fig 6: Front elevation with level datums.

2.4 Room Area Comparison

The largest programmed spaces on campus by floor area:



LIB

Library

258 m²

CRT

Courtyard

237 m²

CNF

Conference

214 m²

TCH

Teachers'

123 m²

LAB

Science Lab

77 m²

PC

Computer Lab

77 m²

MUS

Music Room

74 m²

CLS

Classroom

49 m²

3. GROUND FLOOR DESIGN

The ground floor hosts primary-level education (Nursery through Class VIII), administrative functions, and the central courtyard. It serves as the main entry level with counter (+450 mm), office (+900 mm), and store (+900 mm) level changes accommodating site topography.

3.1 Room Inventory

Space	Dimensions	Area	Function
Classrooms (Nursery - VIII)	7 m × 7 m	49 m ² each	Primary & middle school sections A & B
Principal's Room	5 m × 10.25 m	51 m ²	Head of institution
Office	7.5 m × 7 m	53 m ²	Administrative staff
Parents' Waiting	4 m × 7 m	28 m ²	Visitor reception
Sick Room	5 m × 7 m	35 m ²	First aid & medical care
Store	-	10.25 m length	Equipment storage
Courtyard + Badminton	21.5 m × 11 m	237 m ²	Recreation at +300 mm
Sports Kits Room	-	-	Athletic equipment

Courtyard Design: The central courtyard at +300 mm elevation hosts an integrated badminton court (11 m × 21.5 m), creating a multi-purpose open space that brings daylight into surrounding doubly loaded corridors and provides recreational infrastructure without consuming additional site area.

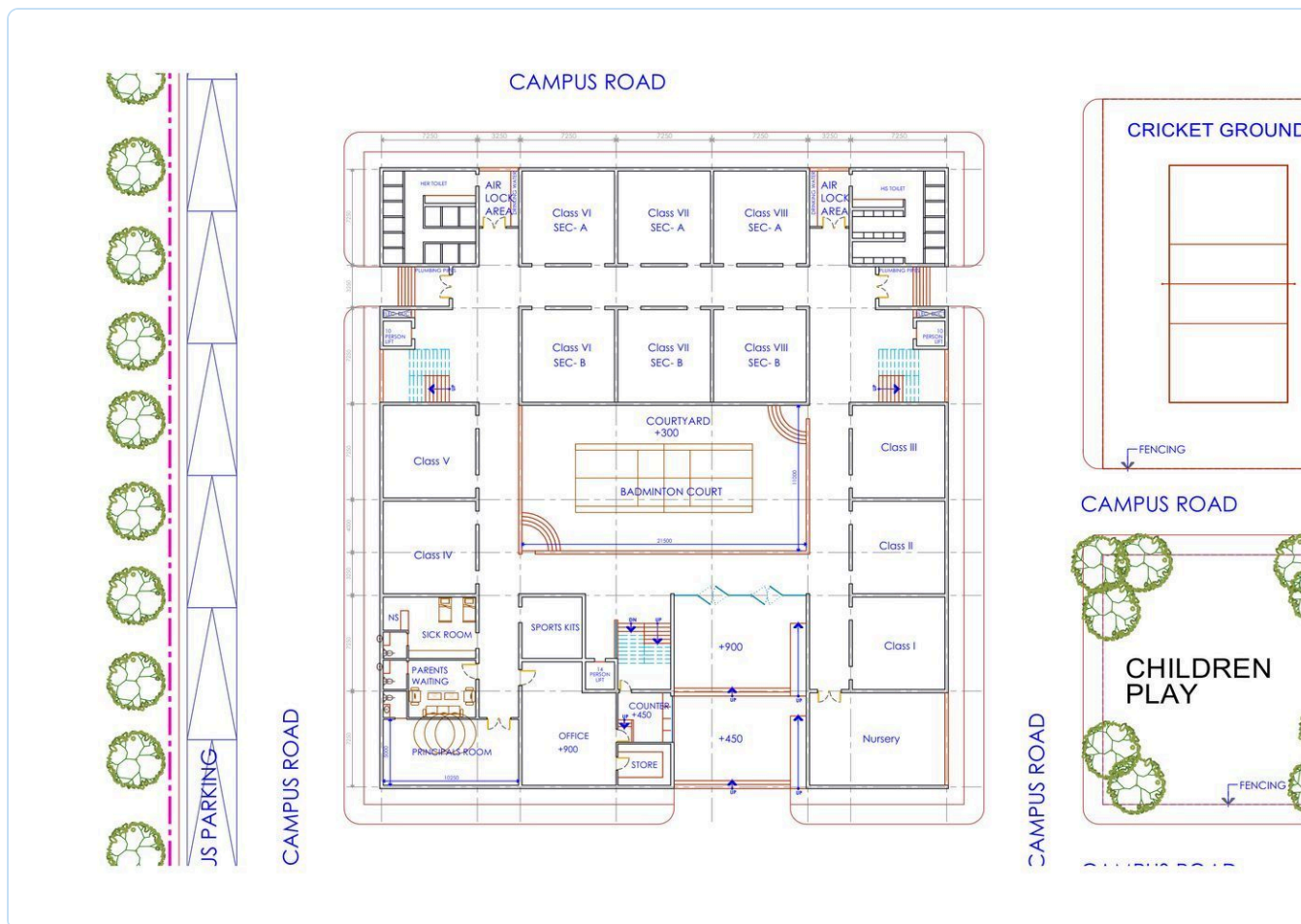


Fig 7: Detailed ground floor plan with dimensional annotations, lift cores, toilet blocks (HIS/HER with air-lock areas), and drinking water stations.

3.2 Vertical Circulation

- **10-person lifts × 2** - positioned at opposite wings for balanced access
- **14-person lift × 1** - adjacent to admin/children's play zone for high-traffic periods
- **Staircases** - at each wing terminus with UP/DN markers; accessible pathways throughout
- **Plumbing & electrical ducts** - routed along corridor spines for service access

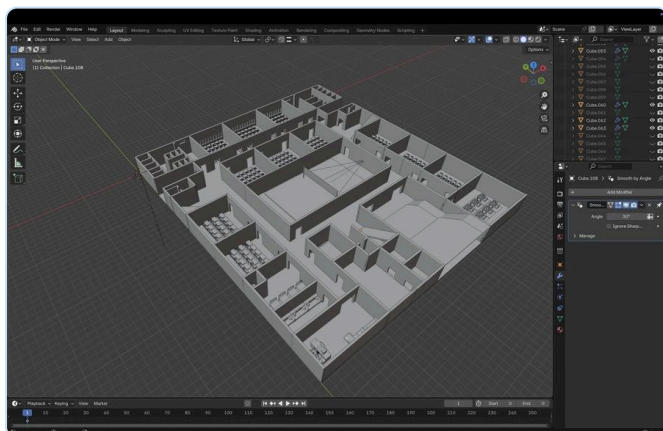


Fig 8: Blender 3D isometric view - ground floor classroom layout with furniture.

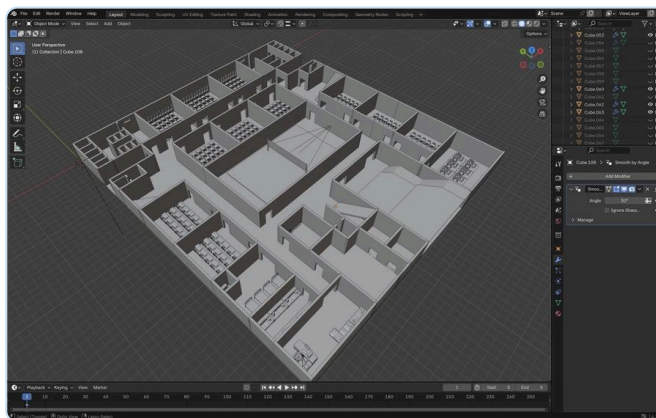


Fig 9: Blender perspective - courtyard and classroom wing relationship.

3.3 Ground Floor Zoning Diagram

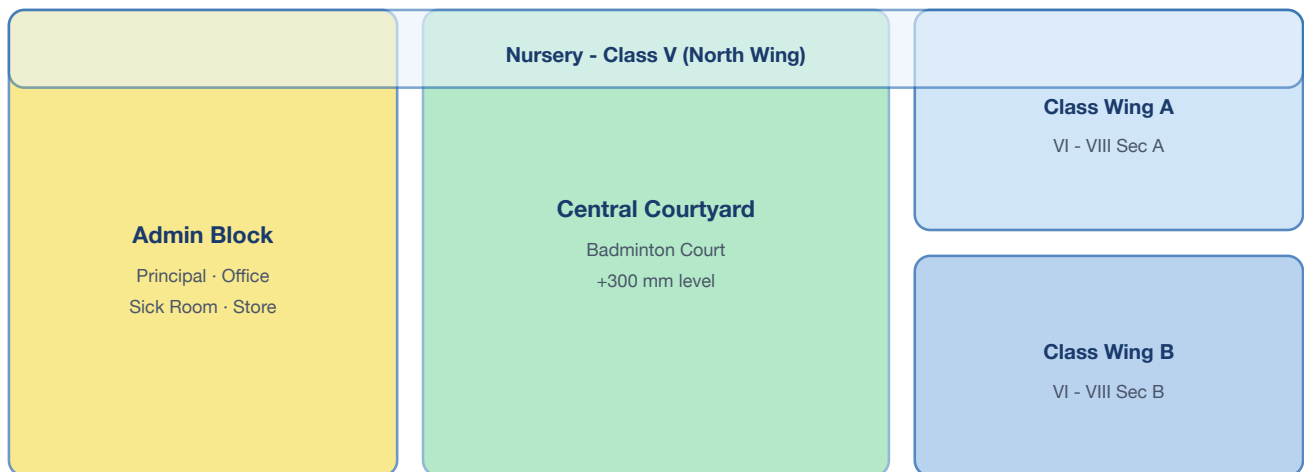


Fig 10: Ground floor zoning - admin block (west), central courtyard, classroom wings (east), and primary classes (north).

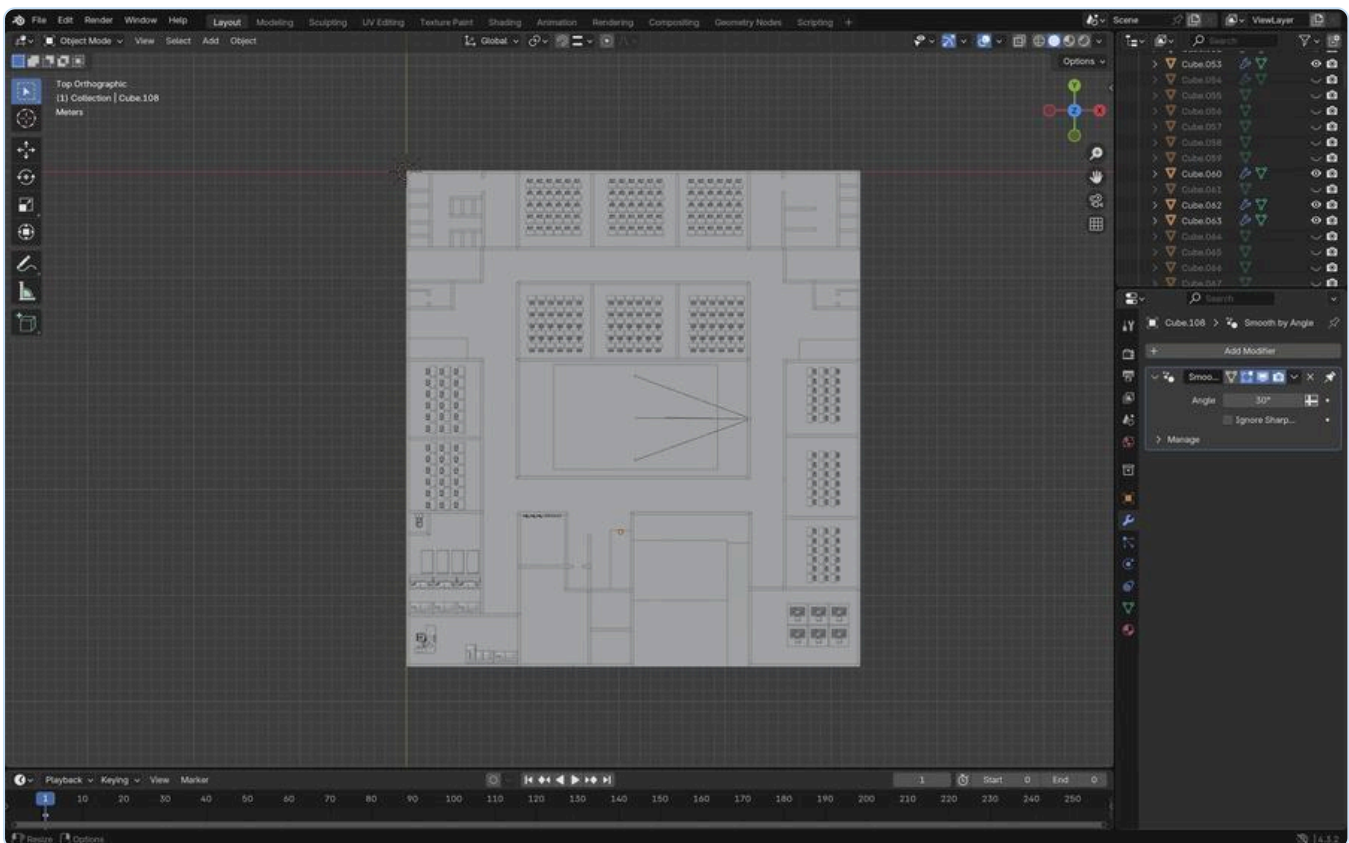


Fig 11: Blender top orthographic view of ground floor plan showing corridor network and room arrangement.

4. FIRST FLOOR DESIGN

The first floor hosts secondary-level classrooms for **Class IX through XII**, organized by section (A, B, C) and academic stream (Science, Commerce, Arts). Faculty facilities and gender-segregated amenities are provided on each wing.

4.1 Classroom & Faculty Layout

Space	Dimensions	Area	Notes
Class IX (Sec A, B, C)	7 m × 7 m	49 m ² each	Three sections
Class X (Sec A, B, C)	7 m × 7 m	49 m ² each	Three sections
Class XI Science / Commerce / Arts	7 m × 7 m	49 m ² each	Stream-wise allocation
Class XII Science / Commerce / Arts	7 m × 7 m	49 m ² each	Stream-wise allocation
Teachers' Lounge	17.5 m × 7 m	123 m ²	Faculty rest & preparation
HIS / HER Toilets	Per wing	-	Separate facilities with air-lock entries
Drinking Water	Per wing	-	Two stations on floor



Fig 12: First floor plan - Class IX-XII with stream-wise allocation and teachers' lounge.

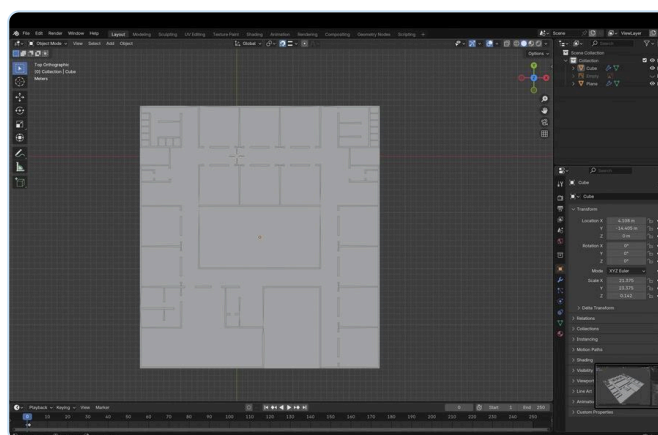


Fig 13: Blender top orthographic view of first floor layout.

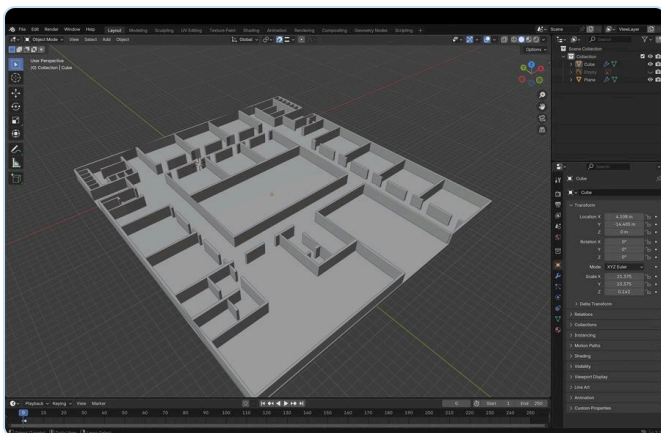


Fig 14: First floor 3D isometric - secondary classroom arrangement.

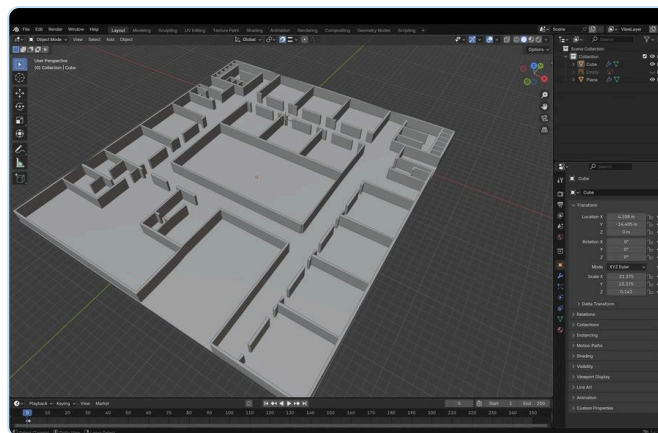


Fig 15: First floor 3D detail - corridor and classroom interface.

Stream Allocation: Class XI and XII are organized by academic stream - Science, Commerce, and Arts - enabling proximity-based scheduling and reducing cross-campus movement for senior students. The teachers' lounge (123 m²) provides dedicated faculty space separate from administrative offices on the ground floor.

5. SECOND FLOOR DESIGN

The second floor is dedicated to **specialized academic and extracurricular spaces** - science laboratories, library, computer lab, music room, and large-format meeting facilities. This vertical separation keeps potentially hazardous lab activities and high-traffic event spaces away from younger students on lower floors.

5.1 Specialized Spaces

Space	Dimensions	Area	Function
Physics Lab	11 m × 7 m	77 m ²	Physics experiments & demonstrations
Chemistry Lab	11 m × 7 m	77 m ²	Chemical experiments with ventilation
Biology Lab	11 m × 7 m	77 m ²	Biological specimens & microscopy
Computer Lab	11 m × 7 m	77 m ²	Modern computing systems
Library + Reading Area	12 m × 21.5 m	258 m ²	Largest single space on campus
Music Room	10.5 m × 7 m	74 m ²	Creative & performing arts
Seminar Room	7 m × 11 m	77 m ²	Events & presentations
Conference Room	15.05 m × 14.25 m	214 m ²	Large meetings & assemblies
Terrace	6.45 m × 7 m	45 m ²	Open-air breakout space

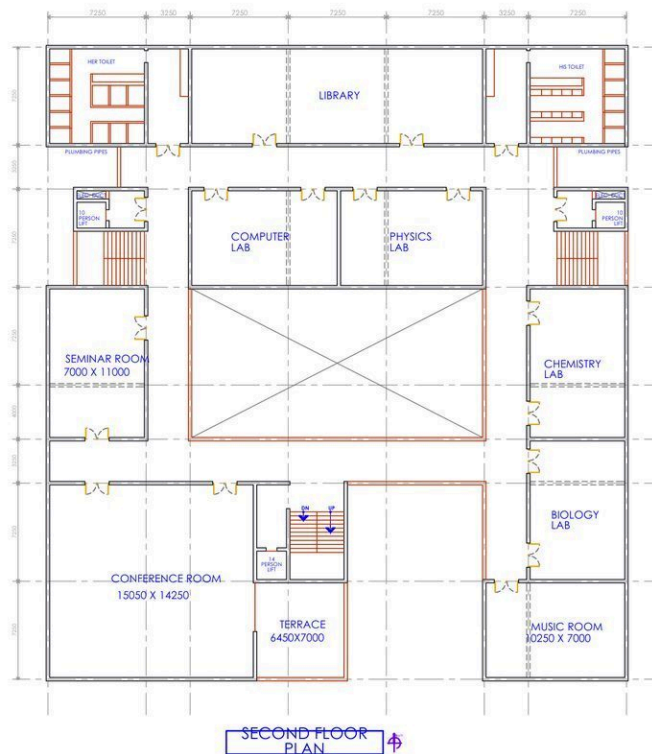


Fig 16: Second floor plan - science labs, library, computer lab, music room, seminar and conference rooms with terrace.

5.2 Lab & Library Zone Diagram

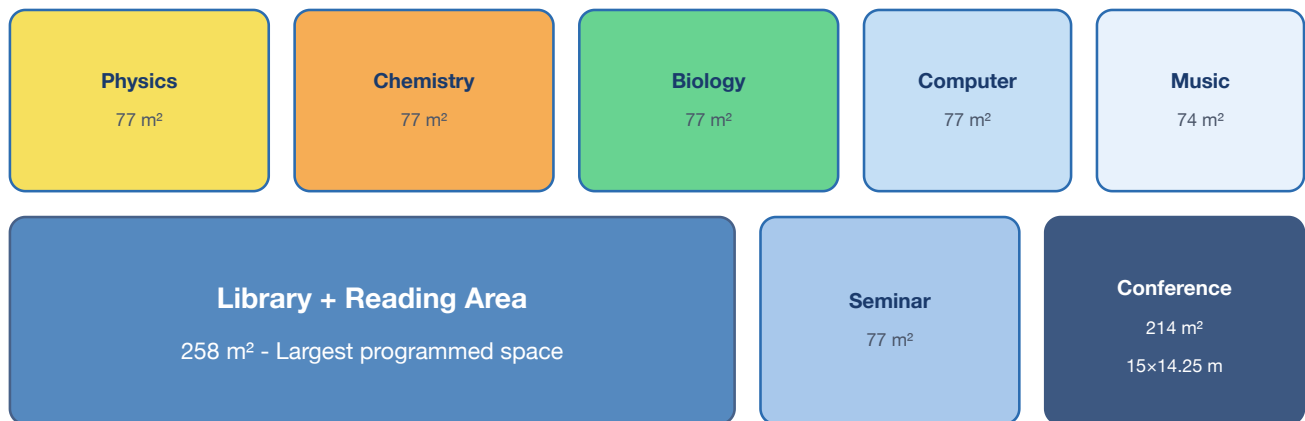


Fig 17: Second floor zoning - color-coded lab blocks, library (central), and meeting facilities.

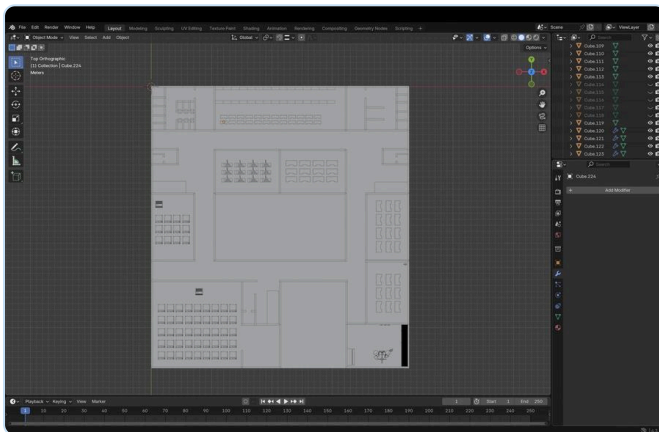


Fig 18: Second floor 3D visualization.

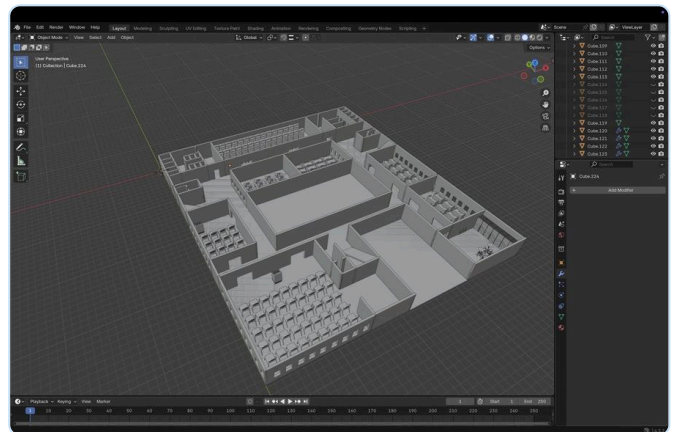


Fig 19: Second floor lab wing detail.

6. FRONT ELEVATION & MASSING

The front elevation showcases the school's **modern design language**: a large entrance with pedestrian pathways and a wheelchair-accessible ramp, spacious parking for staff and visitors, green landscaping for a pleasant environment, and wide windows for natural light and ventilation.

6.1 Level Datum Table

Level	Datum	Description
Road Level	± 0 mm	Campus access road datum
Plinth Level	+900 mm	Building base above road
Ground Floor	First occupied level	Nursery - Class VIII, admin
First Floor	+3600 mm above GF slab	Class IX - XII
Second Floor	+3600 mm above FF slab	Labs, library, conference
Roof Level	+3600 mm above SF slab	Terrace and roof slab



Fig 20: Front elevation - BGA School with floor level annotations and overall building height.

6.2 Campus Massing & 3D Shell

The campus massing model developed in Blender incorporates landscaping elements, building shell geometry, and site context. The building presents a balanced horizontal mass with the central courtyard creating a void that reduces visual bulk while improving daylight penetration.

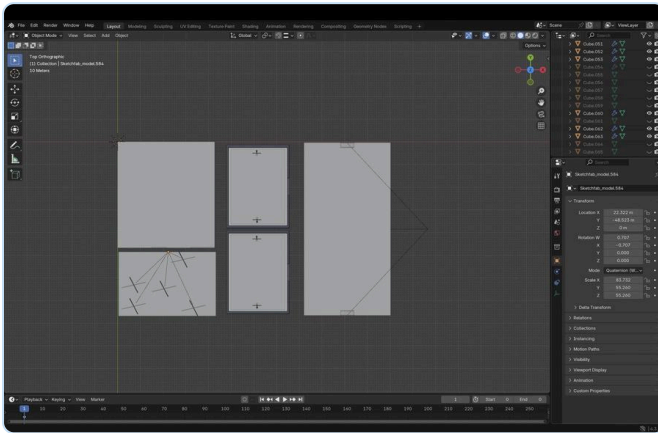


Fig 21: Campus massing - top orthographic site layout.

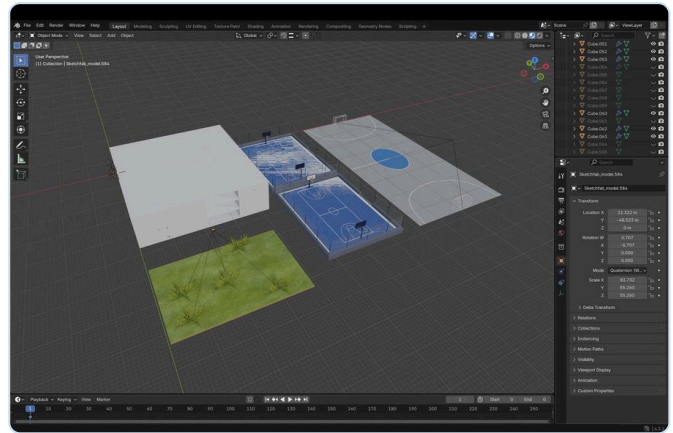


Fig 22: Building shell and site elements.

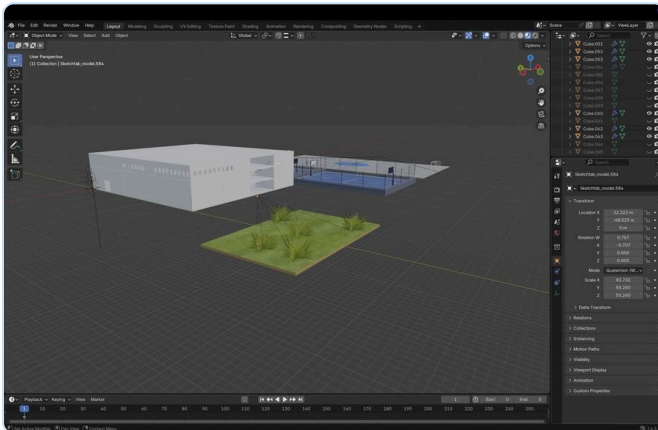


Fig 23: Landscaping and vegetation placement.

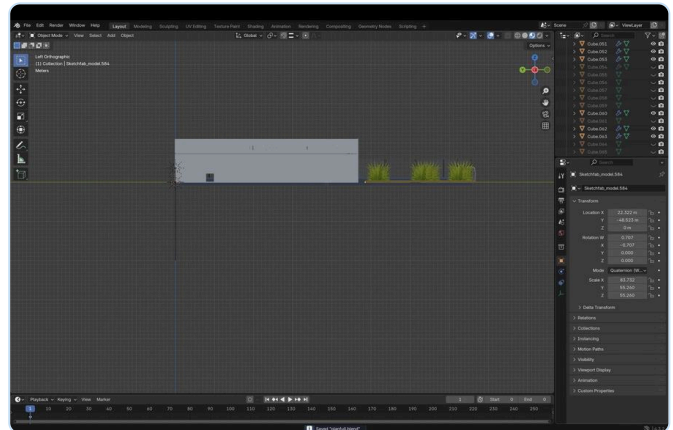


Fig 24: Building facade and ground plane relationship.

7. STRUCTURAL & MATERIAL CONSIDERATIONS

Foundation	Envelope	Finishes
RCC: Primary load-bearing with earthquake resistance. Piles / Deep Footings: For weak soil conditions. SRC: Steel-reinforced concrete for enhanced strength.	Brick Masonry: Cost-effective with plastering. AAC Blocks: Lightweight, fire-resistant insulation. Precast Panels: Reduced construction time.	Vitrified Tiles: High-traffic corridors. Terrazzo: Public spaces and hallways. Rubber Flooring: Play areas and special needs rooms.

7.1 Foundation

Reinforced Concrete (RCC) provides strong load-bearing capacity and resistance against earthquakes. **Piles or deep footings** are specified for areas with weak soil conditions. **Steel Reinforced Concrete (SRC)** incorporates additional steel for enhanced structural strength in critical zones.

7.2 Walls

Brick masonry with cement mortar and plastering serves as the primary wall system. **Concrete blocks** offer better thermal insulation. **Autoclaved Aerated Concrete (AAC) blocks** provide lightweight, fire-resistant construction. **Glass curtain walls** are proposed for modern facade sections to maximize natural light.

7.3 Roof

RCC slab with waterproofing for durability. **Green roofs** (vegetation on waterproof layers) are proposed for insulation and sustainability. **Polycarbonate panels** allow natural light in selected areas while being impact-resistant.

7.4 Flooring

- **Vitrified tiles** - non-slip, easy to maintain, high-traffic corridors and classrooms
- **Ceramic tiles** - restrooms and laboratories (water-resistant)
- **Terrazzo flooring** - corridors and public spaces
- **Vinyl flooring** - classrooms and play areas (soft, slip-resistant)
- **Rubber flooring** - special needs classrooms and playgrounds
- **Polished concrete** - hallways and auditoriums (cost-effective, durable)

7.5 Doors & Windows

- **Aluminium frames with glass panels** - lightweight, rust-resistant, natural light
- **Wooden doors** - classrooms and offices (insulation)
- **Steel doors** - emergency exits (fire resistance)
- **UPVC windows** - energy-efficient, low maintenance
- **Tempered / laminated glass** - safety glazing throughout

7.6 Structural Reinforcement

Steel beams and columns provide structural support for the G+2 frame. **Reinforced concrete frames** are designed for seismic and wind loads. **Bracing systems** (cross-bracing, shear walls) improve earthquake resistance. **Fiber-Reinforced Polymer (FRP)** reinforcements offer lightweight supplemental support.

Sustainability Integration: Solar panels are provisioned for sustainable energy generation. Cork boards and soft boards serve as classroom bulletin surfaces. Fireproof and water-resistant paints ensure safety and durability across all interior surfaces.

8. SAFETY, SUSTAINABILITY & ACCESSIBILITY

The school incorporates comprehensive safety, environmental, and accessibility considerations throughout the design - from fire-rated G+2 construction to wheelchair ramps and rainwater harvesting.

FIRE

Emergency Exits & Fire Safety

Clearly marked fire exits and fire extinguishers on each floor. Steel doors at emergency egress points. G+2 construction in fire-resisting materials.

ACC

Disabled Accessibility

Ramps at main entrance, elevators (10-person × 2, 14-person × 1), and wide doorways for wheelchair access throughout the building.

AIR

Natural Ventilation & Lighting

Maximized through large windows, courtyard design, and doubly loaded corridor typology. Wide windows on front elevation.

PWR

Energy Efficiency

LED lighting throughout, extensive natural daylighting, and provision for rooftop solar panel installation.

H2O

Water Management

Rainwater harvesting system and proper drainage planning. Drinking water stations on each floor wing.

ECO

Green Campus

Landscape gardens, green area buffers, green roof option, and fenced children's play zones for safe outdoor activity.



Fig 25: Safety and sustainability features icon grid.

9. 3D VISUALIZATION & MODELING

The project team developed comprehensive **3D visualizations using Blender 4.3.2**, translating 2D CAD floor plans into spatial models that validate room proportions, circulation paths, and furniture layouts.

9.1 Modeling Workflow

- **2D to 3D:** Floor plan PDFs imported as reference images; walls extruded from traced outlines
- **Modular construction:** Building composed of primitive mesh objects with Smooth-by-Angle modifiers (30° threshold)
- **Furniture placement:** Desk-and-chair arrays in classrooms; lab benches in science rooms
- **Multi-floor assembly:** Ground, first, and second floors modeled as separate collections
- **Site context:** Campus massing with landscaping imported via Sketchfab model integration

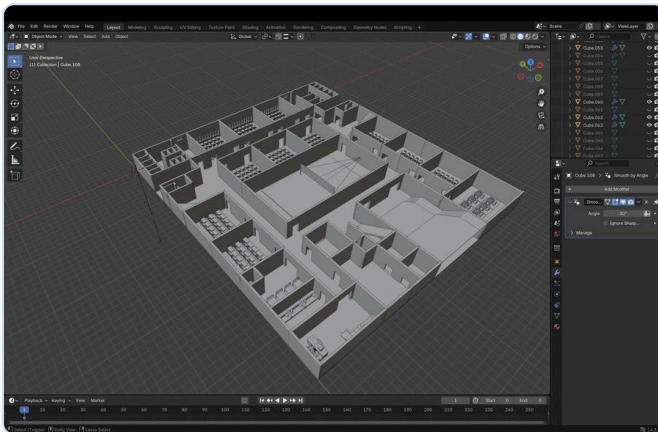


Fig 26: Ground floor - 2D plan to 3D model with furniture.

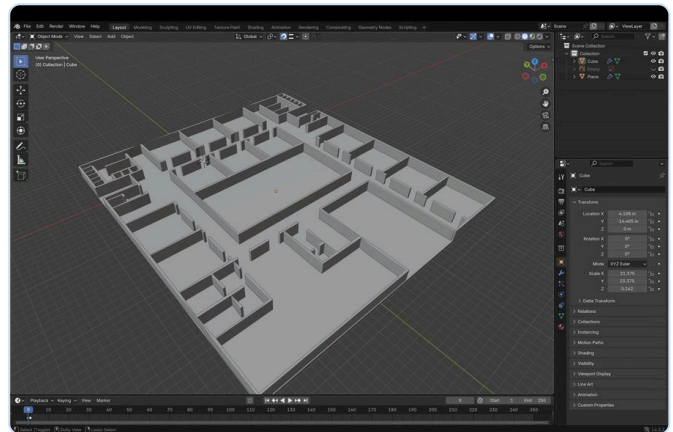


Fig 27: First floor - 3D classroom validation.

9.2 Video Walkthroughs

Animated walkthrough videos were produced for the ground floor and second floor:

- **Ground Floor:** ground/groundfloorglimpse.mp4
- **Second Floor:** secondfloor/secondfloorglimpse.mp4

Source Blender file: firstfloor/floorplan_3d.blend

10. CONCLUSION & KEY LEARNINGS

The **BGA School Building Project** is meticulously designed for optimal academic and extracurricular growth. The well-planned G+2 structure ensures smooth operations, safety, and a dynamic learning experience across all age groups from Nursery through Class XII.

10.1 Design Summary

- **G+2 building** with doubly loaded corridors and central courtyard
- **Complete sports infrastructure** - cricket, football, basketball, badminton, children's play
- **Segregated parking** for school buses, staff, and visitors
- **Specialized second floor** - science labs, library (258 m²), computer lab, music room, conference room (214 m²)
- **Accessibility** - ramps, three elevators, wide corridors and doorways
- **Sustainability** - natural ventilation, solar provision, rainwater harvesting, green landscaping

10.2 Key Learnings

Floor Planning	Technical Documentation	Collaboration
Understanding spatial optimization for classroom layout, office spaces, and movement corridors across multiple building levels.	Creating detailed project plans, CAD drawings, dimensional annotations, and comprehensive architectural reports.	Experience in team collaboration, 3D modeling in Blender, and presenting technical information to diverse audiences.

Final Note: This project demonstrates the integration of architectural planning, structural material selection, sustainability principles, and 3D visualization into a cohesive campus design - providing a foundation for real-world school infrastructure development.

BGA School Building Project

CES Project - Group 3

Architectural Design Report

Floor Plans: schoolplanbga.pdf · groundfloorzoombg.pdf · firstfloorbg.pdf · secondfloorbg.pdf

3D Models: floorplan_3d.blend · Blender 4.3.2 renders

Walkthroughs: groundfloorglimpse.mp4 · secondfloorglimpse.mp4

Presentation: School Building Project.pptx

Generated: June 2025

© CES Project Group 3 - All Rights Reserved